

# Thread Scaling Summary: SMT Efficiency Analysis

## Single Precision - 2000 Time Steps (Values in MLUPs/s)

| Grid Size | 1 Thread | 12 (Physical) | 24 (SMT) | Speedup (12→24) | Insight                        |
|-----------|----------|---------------|----------|-----------------|--------------------------------|
| 1000×1000 | 105      | 693           | 1206     | 1.74x           | Excellent SMT benefit          |
| 2000×2000 | 101      | 607           | 871      | 1.44x           | Good SMT benefit               |
| 3000×3000 | 103      | 515           | 593      | 1.15x           | Moderate SMT benefit           |
| 4000×4000 | 102      | 520           | 523      | 1.01x           | SMT provides almost no benefit |